

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON
ACADEMIC MANAGEMENT SYSTEM

P R O J E C T R E P O R T

Submitted by

SWATHI N

25ITR151

JAVA ASSIGNMENT

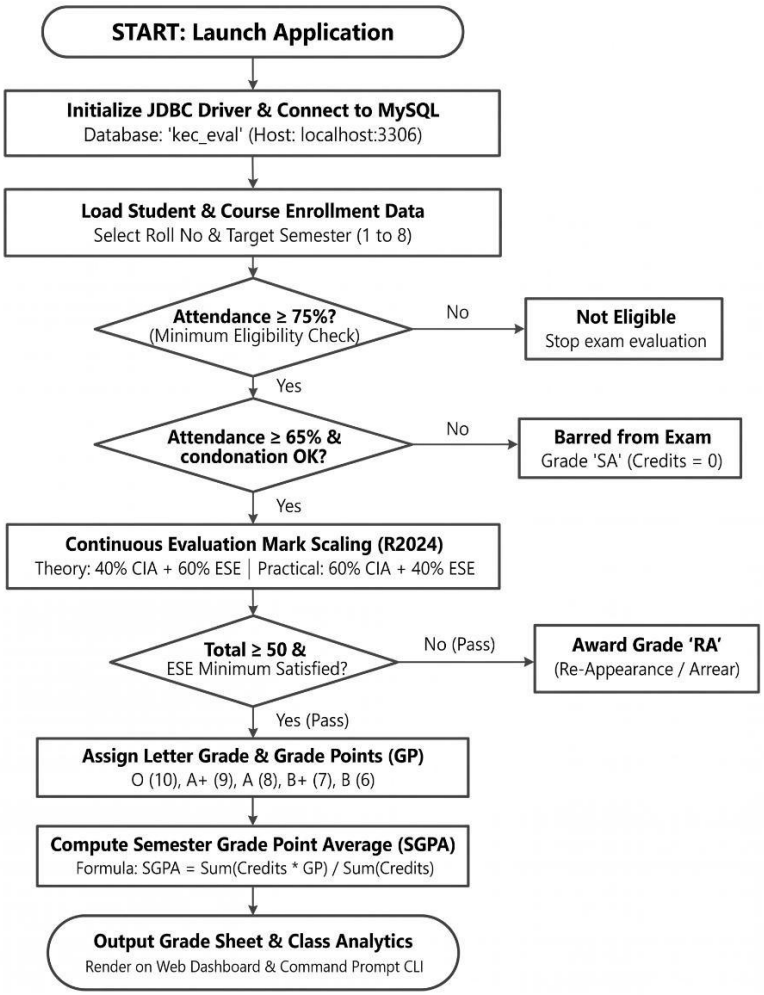
AIM:

Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

FLOW CHART:



PROCEDURE:

- Load the MySQL Connector/J driver to enable communication between the Java application and the MySQL database.

- Establish a connection to the kec_eval database running on the local MySQL server using DriverManager.getConnection().

- Store and retrieve student details, including roll number, name, department, and semester, along with course information such as course code, title, credits, and course type.

- Enroll students in semester courses, record the number of classes held and attended, and calculate the attendance percentage.

- Verify examination eligibility based on attendance percentage. Students with 75% or above are eligible, those with 65%–74% require medical condonation verification, and those below 65% are marked as Shortage of Attendance (SA).

- Record CIA-1, CIA-2, and assignment marks, and calculate the scaled internal assessment marks according to the specified theory and practical course weightages.

- Record end-semester examination marks out of 100, verify the minimum passing requirements, and scale the marks according to the applicable theory and practical .

- Calculate the total marks by combining scaled CIA and ESE marks. Assign letter grades and grade points according to the specified grading scale, marking students as RA if they fail the examination or score below the required passing mark.

- Generate student grade sheets, attendance shortage reports, and department-wise pass percentage reports. Display the results through the interactive web dashboard and the command prompt console**.**

DATABASE SCHEMA:

1.STUDENTS TABLE:

```
mysql> DESCRIBE students;
+------------------+--------------+------+-----+-------------------+-------------------+
| Field            | Type         | Null | Key | Default           | Extra             |
+------------------+--------------+------+-----+-------------------+-------------------+
| student_id       | bigint       | NO   | PRI | NULL              | auto_increment    |
| roll_no          | varchar(20)  | NO   | UNI | NULL              |                   |
| student_name     | varchar(100) | NO   |     | NULL              |                   |
| department       | varchar(20)  | NO   |     | NULL              |                   |
| current_semester | tinyint      | NO   |     | NULL              |                   |
| created_at       | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------------+--------------+------+-----+-------------------+-------------------+
```

2. COURSES TABLE:

```
mysql> DESCRIBE courses;
+--------------+-----------------------------------------+------+-----+---------+----------------+
| Field        | Type                                    | Null | Key | Default | Extra          |
+--------------+-----------------------------------------+------+-----+---------+----------------+
| course_id    | bigint                                  | NO   | PRI | NULL    | auto_increment |
| course_code  | varchar(20)                             | NO   | UNI | NULL    |                |
| course_title | varchar(150)                            | NO   |     | NULL    |                |
| credits      | tinyint                                 | NO   |     | NULL    |                |
| course_type  | enum('THEORY','PRACTICAL','EMBEDDED')   | NO   |     | NULL    |                |
+--------------+-----------------------------------------+------+-----+---------+----------------+
```

3. ENROLLMENTS TABLE:

```
mysql> describe enrollments;
+----------------------+------------+------+-----+-------------------+-------------------+
| Field                | Type       | Null | Key | Default           | Extra             |
+----------------------+------------+------+-----+-------------------+-------------------+
| enrollment_id        | bigint     | NO   | PRI | NULL              | auto_increment    |
| student_id           | bigint     | NO   | MUL | NULL              |                   |
| course_id            | bigint     | NO   | MUL | NULL              |                   |
| semester             | tinyint    | NO   |     | NULL              |                   |
| classes_held         | int        | NO   |     | 0                 |                   |
| classes_attended     | int        | NO   |     | 0                 |                   |
| condonation_approved | tinyint(1) | NO   |     | 0                 |                   |
| enrolled_at          | timestamp  | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+----------------------+------------+------+-----+-------------------+-------------------+
```

4. ASSESSMENT MARKS TABLE:

```
mysql> describe assessment_marks;
+-----------------+-------------+------+-----+-------------------+-----------------------------------------------+
| Field           | Type        | Null | Key | Default           | Extra                                         |
+-----------------+-------------+------+-----+-------------------+-----------------------------------------------+
| mark_id         | bigint      | NO   | PRI | NULL              | auto_increment                                |
| enrollment_id   | bigint      | NO   | UNI | NULL              |                                               |
| cia1            | decimal(5,2)| YES  |     | NULL              |                                               |
| cia2            | decimal(5,2)| YES  |     | NULL              |                                               |
| assignment_mark | decimal(5,2)| YES  |     | NULL              |                                               |
| quiz_mark       | decimal(5,2)| YES  |     | NULL              |                                               |
| lab_mark        | decimal(5,2)| YES  |     | NULL              |                                               |
| ese_mark        | decimal(5,2)| YES  |     | NULL              |                                               |
| ese_absent      | tinyint(1)  | NO   |     | 0                 |                                               |
| updated_at      | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED on update CURRENT_TIMESTAMP |
+-----------------+-------------+------+-----+-------------------+-----------------------------------------------+
```

SOURCE CODE:

MAINAPP.JAVA:

```
import java.sql.SQLException;
import java.util.Scanner;

/**
 * MainApp - Menu-driven CLI interface for Kongu Engineering College (Autonomous)
 * Regulation 2024 (R2024) Student Academic & Continuous Evaluation System.
 */
public class MainApp {

    private static final Scanner scanner = new Scanner(System.in);
    private static final StudentDAO studentDAO = new StudentDAO();
    private static final AcademicDAO academicDAO = new AcademicDAO();

    public static void main(String[] args) {
        printBanner();

        if (!DBConnection.testConnection()) {
            System.err.println("[CRITICAL] Could not establish connection to MySQL database kec_r2024.");
            System.err.println("Please ensure MySQL service is running and schema is loaded via sql/kec_schema.sql.");
            return;
```

```
        }
        System.out.println("[SUCCESS] Connected to MySQL database (kec_r2024)
successfully.\n");

        boolean running = true;
        while (running) {
            printMainMenu();

            int choice = readInt("Enter your choice: ");
            try {
                switch (choice) {
                    case 1 -> handleAddStudent();
                    case 2 -> handleAddCourse();
                    case 3 -> handleEnrollment();
                    case 4 -> handleUpdateAttendance();
                    case 5 -> handleIncrementalMarksMenu();
                    case 6 -> handleStudentMarkSheet();
                    case 7 -> handleAttendanceDefaulters();
                    case 8 -> handleClassSummary();
                    case 9 -> handleViewCatalog();
                    case 0 -> {
                        System.out.println("\nThank you for using KEC R2024 Academic Evaluation
System. Goodbye!");
                        running = false;
                    }
                    default -> System.out.println("\n[ERROR] Invalid menu choice. Please select
from 0 to 9.");
                }
            } catch (SQLException e) {
                System.err.println("\n[DATABASE ERROR] " + e.getMessage());
            } catch (IllegalArgumentException e) {
                System.err.println("\n[VALIDATION ERROR] " + e.getMessage());
            } catch (Exception e) {
                System.err.println("\n[ERROR] Unexpected error: " + e.getMessage());
            }
        }
```

```
    }
    private static void printBanner() {
        System.out.println("================================================
=========================================");
        System.out.println("          KONGU ENGINEERING COLLEGE (AUTONOMOUS) -
REGULATION 2024 (R2024)            ");
        System.out.println("              Student Academic & Continuous Evaluation System
(JDBC)                ");
        System.out.println("================================================
=========================================");
    }
    private static void printMainMenu() {
        System.out.println("\n--------------------------------------------------------------------------------
----------");
        System.out.println("                              MAIN MENU                                ");
        System.out.println("--------------------------------------------------------------------------------
-------");
        System.out.println("  1. Student Registration (Add New Student)");
        System.out.println("  2. Course Catalog (Add New Course: Theory / Practical /
Embedded)");
        System.out.println("  3. Course Enrollment & Attendance Tracking");
        System.out.println("  4. Update Attendance & Record Medical Condonation Approval");
        System.out.println("  5. Phased / Incremental Marks Entry ('Then and Now': CIA1,
CIA2, Lab, ESE, AB)");
        System.out.println("  6. Generate Student Semester Mark Sheet (Formal Grade Report &
SGPA)");
        System.out.println("  7. Generate Attendance Defaulter Report (Grade SA Flagging)");
        System.out.println("  8. Generate Class Performance & Result Summary (Dept Pass/Fail
& Class Avg SGPA)");
        System.out.println("  9. View Registered Students & Course Catalog");
        System.out.println("  0. Exit System");
        System.out.println("--------------------------------------------------------------------------------
-------");
    }
    private static void handleAddStudent() throws SQLException {
        System.out.println("\n--- [1] STUDENT REGISTRATION ---");
```

```
        String rollNo = readString("Enter Roll Number (e.g. 24ITR001): ");
        String name = readString("Enter Student Full Name: ");
        String dept = readString("Enter Department (CSE, IT, ECE, MECH, etc.): ");
        int sem = readInt("Enter Current Semester (1-8): ");

        boolean added = studentDAO.addStudent(rollNo, name, dept, sem);
        if (added) {
            System.out.println("[SUCCESS] Student registered successfully: " +
rollNo.toUpperCase() + " - " + name);
        }
    }
    private static void handleAddCourse() throws SQLException {
        System.out.println("\n--- [2] COURSE CATALOG ENTRY ---");
        String code = readString("Enter Course Code (e.g. 24ITT31): ");
        String title = readString("Enter Course Title: ");
        int credits = readInt("Enter Course Credits (1-10): ");
        System.out.println("Select Course Type:");
        System.out.println("  1. THEORY (CIA: 40%, ESE: 60%, ESE Min: 45%)");
        System.out.println("  2. PRACTICAL (CIA: 60%, ESE: 40%, ESE Min: 50%)");
        System.out.println("  3. EMBEDDED (CIA: 50%, ESE: 50%, ESE Min: 45%)");
        int typeChoice = readInt("Choose Course Type (1-3): ");
        String courseType = switch (typeChoice) {
            case 1 -> "THEORY";
            case 2 -> "PRACTICAL";
            case 3 -> "EMBEDDED";
            default -> throw new IllegalArgumentException("Invalid course type option.");

        };

        boolean added = studentDAO.addCourse(code, title, credits, courseType);
        if (added) {
            System.out.println("[SUCCESS] Course added successfully: [" + code.toUpperCase()
+ "] " + title + " (" + courseType + ", " + credits + " Credits)");
        }
    }
    private static void handleEnrollment() throws SQLException {
```

```
        System.out.println("\n--- [3] COURSE ENROLLMENT & ATTENDANCE
TRACKING ---");
        String rollNo = readString("Enter Student Roll Number: ");
        String courseCode = readString("Enter Course Code: ");
        int semester = readInt("Enter Semester: ");
        int held = readInt("Enter Total Classes Held: ");
        int attended = readInt("Enter Classes Attended: ");
        boolean condoned = readYesNo("Is Medical Condonation Approved? (y/n): ");
        academicDAO.enroll(rollNo, courseCode, semester, held, attended, condoned);
        System.out.println("[SUCCESS] Student " + rollNo.toUpperCase() + " successfully
enrolled in " + courseCode.toUpperCase() + ".");
    }

    private static void handleUpdateAttendance() throws SQLException {
        System.out.println("\n--- [4] UPDATE ATTENDANCE & CONDONATION ---");
        String rollNo = readString("Enter Student Roll Number: ");
        String courseCode = readString("Enter Course Code: ");
        int semester = readInt("Enter Semester: ");
        int held = readInt("Enter Updated Total Classes Held: ");
        int attended = readInt("Enter Updated Classes Attended: ");
        boolean condoned = readYesNo("Is Medical Condonation Approved? (y/n): ");

        academicDAO.updateAttendance(rollNo, courseCode, semester, held, attended,
condoned);
        double pct = (held > 0) ? (attended * 100.0) / held : 0.0;
        System.out.printf("[SUCCESS] Attendance updated: %d/%d (%.2f%%). Condonation
Approved: %s%n",
                attended, held, pct, condoned ? "YES" : "NO");
    }

    private static void handleIncrementalMarksMenu() throws SQLException {
        boolean back = false;
        while (!back) {
            System.out.println("\n----------------------------------------------------------------------------
-------------");
```

```
        System.out.println("               INCREMENTAL MARKS ENTRY - 'THEN AND
NOW' PHASES                        ");
        System.out.println("-------------------------------------------------------------------------------
----------");
        System.out.println("  1. Phase 1 (Mid-Semester)      : Enter CIA Test 1 (0-100) & CIA
Test 2 (0-100)");
        System.out.println("  2. Phase 2 (End of Semester)   : Enter Assignments / Quizzes /
Lab Evaluations (0-100)");
        System.out.println("  3. Phase 3 (Post Semester)     : Enter ESE Marks (0-100) or Mark
Absence (AB)");
        System.out.println("  4. Comprehensive / Bulk Entry  : Update Any / All Assessment
Components");
        System.out.println("  0. Back to Main Menu");
        System.out.println("-------------------------------------------------------------------------------
----------");

        int phase = readInt("Select Marks Entry Option: ");
        switch (phase) {
            case 1 -> handlePhase1Marks();
            case 2 -> handlePhase2Marks();
            case 3 -> handlePhase3Marks();
            case 4 -> handleComprehensiveMarks();
            case 0 -> back = true;
            default -> System.out.println("[ERROR] Invalid choice. Choose 0 to 4.");
        }
      }
   }

   private static void handlePhase1Marks() throws SQLException {
      System.out.println("\n>>> PHASE 1 (MID-SEMESTER): CIA TEST 1 & CIA TEST
2");
      String rollNo = readString("Enter Roll Number: ");
      String courseCode = readString("Enter Course Code: ");
      int semester = readInt("Enter Semester: ");

      // Show "Then"
```

```
        academicDAO.displayExistingMarks(rollNo, courseCode, semester);

        // Take "Now"
        System.out.println("\n[NOW - Enter New/Updated Phase 1 Marks (Leave blank to keep existing)]");
        Double cia1 = readOptionalDouble("Enter CIA Test 1 Mark (0-100): ");
        Double cia2 = readOptionalDouble("Enter CIA Test 2 Mark (0-100): ");

        academicDAO.savePhase1Marks(rollNo, courseCode, semester, cia1, cia2);
        System.out.println("[SUCCESS] Phase 1 Marks successfully recorded for " + rollNo.toUpperCase() + " in " + courseCode.toUpperCase() + ".");
    }

    private static void handlePhase2Marks() throws SQLException {

        System.out.println("\n>>> PHASE 2 (END OF SEMESTER CIA): CONTINUOUS ASSESSMENTS");
        String rollNo = readString("Enter Roll Number: ");
        String courseCode = readString("Enter Course Code: ");
        int semester = readInt("Enter Semester: ");

        // Show "Then"
        academicDAO.displayExistingMarks(rollNo, courseCode, semester);

        // Take "Now"
        System.out.println("\n[NOW - Enter New/Updated Continuous Assessment Marks (Leave blank to skip)]");
        Double assignment = readOptionalDouble("Enter Assignment Mark (0-100): ");
        Double quiz = readOptionalDouble("Enter Quiz Mark (0-100): ");
        Double lab = readOptionalDouble("Enter Lab Evaluation Mark (0-100): ");

        academicDAO.savePhase2Marks(rollNo, courseCode, semester, assignment, quiz, lab);
        System.out.println("[SUCCESS] Phase 2 Marks successfully recorded for " + rollNo.toUpperCase() + " in " + courseCode.toUpperCase() + ".");
    }
```

```
    private static void handlePhase3Marks() throws SQLException {
        System.out.println("\n>>> PHASE 3 (POST SEMESTER): END SEMESTER
EXAMINATION (ESE)");
        String rollNo = readString("Enter Roll Number: ");
        String courseCode = readString("Enter Course Code: ");
        int semester = readInt("Enter Semester: ");

        // Show "Then"
        academicDAO.displayExistingMarks(rollNo, courseCode, semester);

        // Take "Now"
        System.out.println("\n[NOW - Enter ESE Mark or Absence]");
        String input = readString("Enter ESE Mark (0-100) or 'AB' for Absent: ");

        boolean isAbsent = input.equalsIgnoreCase("AB");
        Double ese = null;
        if (!isAbsent && !input.isBlank()) {
            ese = Double.parseDouble(input.trim());
        }

        academicDAO.savePhase3Marks(rollNo, courseCode, semester, ese, isAbsent);
        System.out.println("[SUCCESS] Phase 3 ESE evaluation successfully recorded for " +
rollNo.toUpperCase() + " in " + courseCode.toUpperCase() + ".");
    }


    private static void handleComprehensiveMarks() throws SQLException {
        System.out.println("\n>>> COMPREHENSIVE ASSESSMENT MARKS ENTRY");
        String rollNo = readString("Enter Roll Number: ");
        String courseCode = readString("Enter Course Code: ");
        int semester = readInt("Enter Semester: ");

        // Show "Then"
        academicDAO.displayExistingMarks(rollNo, courseCode, semester);
        System.out.println("\n[NOW - Enter assessment components (Press Enter to keep
unchanged)]");
```

```
        Double cia1 = readOptionalDouble("CIA Test 1 (0-100): ");
        Double cia2 = readOptionalDouble("CIA Test 2 (0-100): ");
        Double assignment = readOptionalDouble("Assignment (0-100): ");
        Double quiz = readOptionalDouble("Quiz (0-100): ");
        Double lab = readOptionalDouble("Lab Evaluation (0-100): ");

        String eseInput = readString("ESE Marks (0-100), 'AB' for Absent, or blank to keep: ");
        boolean eseProvided = !eseInput.isBlank();
        boolean isAbsent = eseInput.equalsIgnoreCase("AB");
        Double ese = (eseInput.isBlank() || isAbsent) ? null :
Double.parseDouble(eseInput.trim());

        academicDAO.saveComprehensiveMarks(rollNo, courseCode, semester, cia1, cia2,
assignment, quiz, lab, ese, eseProvided, isAbsent);
        System.out.println("[SUCCESS] Assessment marks updated successfully.");
    }

    private static void handleStudentMarkSheet() throws SQLException {
        System.out.println("\n--- [6] STUDENT SEMESTER MARK SHEET ---");
        String rollNo = readString("Enter Student Roll Number (e.g. 24ITR001): ");
        int semester = readInt("Enter Semester (1-8): ");
        academicDAO.printStudentMarkSheet(rollNo, semester);
    }

    private static void handleAttendanceDefaulters() throws SQLException {
        academicDAO.generateAttendanceDefaulterReport();
    }

    private static void handleClassSummary() throws SQLException {
        System.out.println("\n--- [8] CLASS PERFORMANCE & RESULT SUMMARY ---");
        int semester = readInt("Enter Semester to analyze (1-8): ");
        academicDAO.generateClassPerformanceSummary(semester);
    }

    private static void handleViewCatalog() throws SQLException {
        studentDAO.listAllStudents();
```

```
        studentDAO.listAllCourses();
    }

    // Helper I/O methods
    private static String readString(String prompt) {
        System.out.print(prompt);
        return scanner.nextLine().trim();
    }

    private static int readInt(String prompt) {
        while (true) {
            System.out.print(prompt);
            String line = scanner.nextLine().trim();
            try {
                return Integer.parseInt(line);
            } catch (NumberFormatException e) {
                System.out.println("[ERROR] Please enter a valid integer.");
            }
        }
    }

    private static Double readOptionalDouble(String prompt) {
        while (true) {
            System.out.print(prompt);
            String line = scanner.nextLine().trim();
            if (line.isEmpty()) {
                return null;
            }
            try {
                double val = Double.parseDouble(line);
                if (val < 0.0 || val > 100.0) {
                    System.out.println("[ERROR] Mark must be between 0.0 and 100.0.");
                    continue;
                }
                return val;
            } catch (NumberFormatException e) {
```

```
                System.out.println("[ERROR] Please enter a valid number or leave blank.");
            }
        }
    }


    private static boolean readYesNo(String prompt) {
        while (true) {
            System.out.print(prompt);
            String line = scanner.nextLine().trim().toLowerCase();
            if (line.equals("y") || line.equals("yes")) return true;
            if (line.equals("n") || line.equals("no")) return false;
            System.out.println("[ERROR] Please answer 'y' or 'n'.");
        }
    }
}
```

DB CONNECTION.JAVA:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

/**
 * DBConnection provides JDBC database connectivity for Kongu Engineering College
(KEC)
 * R2024 Academic Evaluation System using MySQL.
 */
public final class DBConnection {
    private static final String URL =
"jdbc:mysql://localhost:3306/kec_r2024?useSSL=false&allowPublicKeyRetrieval=true&serv
erTimezone=Asia/Kolkata";
    private static final String USER = "root";
    private static final String PASSWORD = "Mysql@12345";

    private DBConnection() {
        // Prevent instantiation
```

```
    }

    static {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            System.err.println("[ERROR] MySQL JDBC Driver not found on classpath: " +
e.getMessage());
        }
    }

    /**
     * Gets a new JDBC database connection.
     * @return Connection object

     * @throws SQLException if a database access error occurs
     */
    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(URL, USER, PASSWORD);
    }

    /**
     * Tests whether the database connection can be established.
     * @return true if successful, false otherwise
     */
    public static boolean testConnection() {
        try (Connection con = getConnection()) {
            return con != null && !con.isClosed();
        } catch (SQLException e) {
            System.err.println("[DBConnection Test Failed] " + e.getMessage());
            return false;
        }
    }
```

OUTPUT SCREENSHOTS:

MAIN MENU:

```
                                MAIN MENU
--------------------------------------------------------------------------------
  1. Student Registration (Add New Student)
  2. Course Catalog (Add New Course: Theory / Practical / Embedded)
  3. Course Enrollment & Attendance Tracking
  4. Update Attendance & Record Medical Condonation Approval
  5. Phased / Incremental Marks Entry ('Then and Now': CIA1, CIA2, Lab, ESE, AB)
  6. Generate Student Semester Mark Sheet (Formal Grade Report & SGPA)
  7. Generate Attendance Defaulter Report (Grade SA Flagging)
  8. Generate Class Performance & Result Summary (Dept Pass/Fail & Class Avg SGPA)
  9. View Registered Students & Course Catalog
  0. Exit System
```

STUDENT & COURSE MANAGEMENT:

1.STUDENT REGISTRATION:

```
--- [1] STUDENT REGISTRATION ---
Enter Roll Number (e.g. 24ITR001): 24ITR056
Enter Student Full Name: Athidhreen
Enter Department (CSE, IT, ECE, MECH, etc.): IT
Enter Current Semester (1-8): 3
[SUCCESS] Student registered successfully: 24ITR056 - Athidhreen
```

2. COURSE CATALOG:

```
--- [2] COURSE CATALOG ENTRY ---
Enter Course Code (e.g. 24ITT31): 24ITT97
Enter Course Title: Web Development
Enter Course Credits (1-10): 3
Select Course Type:
  1. THEORY (CIA: 40%, ESE: 60%, ESE Min: 45%)
  2. PRACTICAL (CIA: 60%, ESE: 40%, ESE Min: 50%)
  3. EMBEDDED (CIA: 50%, ESE: 50%, ESE Min: 45%)
Choose Course Type (1-3): 2
[SUCCESS] Course added successfully: [24ITT97] Web Development (PRACTICAL, 3 Credits)
```

3. COURSE ENROLLMENT & ATTENDANCE TRACKING

```
--- [3] COURSE ENROLLMENT & ATTENDANCE TRACKING ---
Enter Student Roll Number: 24ITR056
Enter Course Code: 24ITT33
Enter Semester: 3
Enter Total Classes Held: 30
Enter Classes Attended: 28
Is Medical Condonation Approved? (y/n): n
[SUCCESS] Student 24ITR056 successfully enrolled in 24ITT33.
```

INCREMENTAL MARKS ENTRY:

1.PHASE 1 (MID-SEMESTER):

```
--------------------------------------------------------------------------------
                INCREMENTAL MARKS ENTRY - 'THEN AND NOW' PHASES
--------------------------------------------------------------------------------
  1. Phase 1 (Mid-Semester)      : Enter CIA Test 1 (0-100) & CIA Test 2 (0-100)
  2. Phase 2 (End of Semester)   : Enter Assignments / Quizzes / Lab Evaluations (0-100)
  3. Phase 3 (Post Semester)     : Enter ESE Marks (0-100) or Mark Absence (AB)
  4. Comprehensive / Bulk Entry  : Update Any / All Assessment Components
  0. Back to Main Menu
--------------------------------------------------------------------------------
Select Marks Entry Option: 1

>>> PHASE 1 (MID-SEMESTER): CIA TEST 1 & CIA TEST 2
Enter Roll Number: 24ITR151
Enter Course Code: 24ITT33
Enter Semester: 3

[THEN - Current Recorded Assessment Marks]
  No marks entered yet (Fresh assessment record).

[NOW - Enter New/Updated Phase 1 Marks (Leave blank to keep existing)]
Enter CIA Test 1 Mark (0-100): 89
Enter CIA Test 2 Mark (0-100): 85
[SUCCESS] Phase 1 Marks successfully recorded for 24ITR151 in 24ITT33.
```

2.PHASE 2 (END OF SEMESTER - CIA):

```
--------------------------------------------------------------------------------
                INCREMENTAL MARKS ENTRY - 'THEN AND NOW' PHASES
--------------------------------------------------------------------------------
  1. Phase 1 (Mid-Semester)      : Enter CIA Test 1 (0-100) & CIA Test 2 (0-100)
  2. Phase 2 (End of Semester)   : Enter Assignments / Quizzes / Lab Evaluations (0-100)
  3. Phase 3 (Post Semester)     : Enter ESE Marks (0-100) or Mark Absence (AB)
  4. Comprehensive / Bulk Entry  : Update Any / All Assessment Components
  0. Back to Main Menu
--------------------------------------------------------------------------------
Select Marks Entry Option: 2

>>> PHASE 2 (END OF SEMESTER CIA): CONTINUOUS ASSESSMENTS
Enter Roll Number: 24ITR151
Enter Course Code: 24ITT33
Enter Semester: 3

[THEN - Current Recorded Assessment Marks]
  CIA Test 1   : 89.00
  CIA Test 2   : 85.00
  Assignment   : --
  Quiz         : --
  Lab Eval     : --
  ESE Mark     : --

[NOW - Enter New/Updated Continuous Assessment Marks (Leave blank to skip)]
Enter Assignment Mark (0-100): 80
Enter Quiz Mark (0-100): 77
Enter Lab Evaluation Mark (0-100): 90
[SUCCESS] Phase 2 Marks successfully recorded for 24ITR151 in 24ITT33.
```

3. PHASE 3 (POST SEMESTER):

```
--------------------------------------------------------------------------------
                INCREMENTAL MARKS ENTRY - 'THEN AND NOW' PHASES
--------------------------------------------------------------------------------
  1. Phase 1 (Mid-Semester)      : Enter CIA Test 1 (0-100) & CIA Test 2 (0-100)
  2. Phase 2 (End of Semester)   : Enter Assignments / Quizzes / Lab Evaluations (0-100)
  3. Phase 3 (Post Semester)     : Enter ESE Marks (0-100) or Mark Absence (AB)
  4. Comprehensive / Bulk Entry  : Update Any / All Assessment Components
  0. Back to Main Menu
--------------------------------------------------------------------------------
Select Marks Entry Option: 3

>>> PHASE 3 (POST SEMESTER): END SEMESTER EXAMINATION (ESE)
Enter Roll Number: 24ITR151
Enter Course Code: 24ITT33
Enter Semester: 3

[THEN - Current Recorded Assessment Marks]
  CIA Test 1   : 89.00
  CIA Test 2   : 85.00
  Assignment   : 80.00
  Quiz         : 77.00
  Lab Eval     : 90.00
  ESE Mark     : --

[NOW - Enter ESE Mark or Absence]
Enter ESE Mark (0-100) or 'AB' for Absent: 88
[SUCCESS] Phase 3 ESE evaluation successfully recorded for 24ITR151 in 24ITT33.
```

ATTENDANCE & GRADING RULES:

1.ATTENDANCE FILTER:

KEC R2024 - ATTENDANCE DEFAULTER REPORT (GRADE SA)
Criteria: Attendance < 65% OR 65-75% without Medical Condonation

Roll No	Student Name	Dept	Sem	Course	Held	Attd	Attend %	Barred Grade
24ITR003	Priya S	IT	Sem3	24ITT31	50	28	56.00%	SA (< 65%)
24ITR003	Priya S	IT	Sem3	24ITT32	60	34	56.67%	SA (< 65%)
24ITR003	Priya S	IT	Sem3	24ITP31	30	18	60.00%	SA (< 65%)
24ITR003	Priya S	IT	Sem3	24ITE31	60	32	53.33%	SA (< 65%)

STUDENT SEMESTER MARK SHEET:

KONGU ENGINEERING COLLEGE (AUTONOMOUS), PERUNDURAI
REGULATION 2024 - GRADE REPORT / MARK SHEET

Roll Number : 24ITR151 Student Name : Swathi Semester : 3
Department : IT Current Sem : Sem 3 Academic Year: 2024-2025

Code	Course Title	Type	Attend %	CIA/100	CIA-Wt	ESE-Wt	Total	Grade	GP	Pass
24ITT33	Data App Development	THEO	93.3%	84.2	33.7	52.8	86.5	A+	9	PASS

--- ACADEMIC PERFORMANCE SUMMARY ---
Total Enrolled Credits : 3
Total Earned Credits : 3
Total Quality Points : 27.00
Semester GPA (SGPA) : 9.00 / 10.00
Equivalent Percentage : 90.00%
Final Semester Status : PASSED - FIRST CLASS WITH DISTINCTION

DEPARTMENT WISE CLASS PERFORMANCE & RESULT SUMMARY:

KEC R2024 - CLASS PERFORMANCE & RESULT SUMMARY (SEMESTER 3)

Department	Total Students	All Passed	Arrears/SA	Pass Pct (%)	Class Avg SGPA	Top SGPA
CSE	2	1	1	50.00%	3.50	7.00 (24CSR138)
ECE	1	1	0	100.00%	8.00	8.00 (24ECR029)
EEE	1	1	0	100.00%	8.00	8.00 (24EER038)
IT	6	5	1	83.33%	7.15	9.46 (24ITR001)

DATABASE:

STUDENTS DATABASE:

REGISTERED STUDENTS LIST

Roll No	Student Name	Department	Semester	Registered Date
24CSR010	Arun V	CSE	Sem 3	2026-10-26
24CSR011	Kavitha P	CSE	Sem 3	2026-10-26
24CSR138	Gowtham	CSE	Sem 3	2026-10-26
24ECR025	Naveen Raj K	ECE	Sem 3	2026-10-26
24ECR029	Dharshini	ECE	Sem 3	2026-10-26
24EER038	Nethra	EEE	Sem 3	2026-10-26
24ITR001	Swathi M	IT	Sem 3	2026-10-26
24ITR002	Dharun Kumar R	IT	Sem 3	2026-10-26
24ITR003	Priya S	IT	Sem 3	2026-10-26
24ITR056	Athidhreen	IT	Sem 3	2026-10-26
24ITR151	Swathi	IT	Sem 3	2026-10-26
24ITR153	Swetha	IT	Sem 3	2026-10-26
24ITT151	Swathi	IT	Sem 3	2026-10-26

COURSES DATABASE:

COURSE CATALOG

Course Code	Course Title	Credits	Course Type
24CST31	Discrete Mathematics	4	THEORY
24ECT31	Signals and Systems	4	THEORY
24ITE31	Object Oriented Programming with Java	4	EMBEDDED
24ITP31	Database Systems Laboratory	2	PRACTICAL
24ITT31	Database Design and Management	3	THEORY
24ITT32	Data Structures and Algorithms	4	THEORY
24ITT33	Data App Development	3	THEORY
24ITT96	Python	3	EMBEDDED
24ITT97	Web Development	3	PRACTICAL
24ITT98	Full stack development	3	PRACTICAL

ENROLLMENTS DATABASE:

enrollment_id	roll_no	course_code	semester	total_classes	attended_classes	is_condoned	created_at
1	24ITR001	24ITT31	3	50	48	0	2026-10-26 18:41:31
2	24ITR001	24ITT32	3	60	56	0	2026-10-26 18:41:31
3	24ITR001	24ITP31	3	30	29	0	2026-10-26 18:41:31
4	24ITR001	24ITE31	3	60	58	0	2026-10-26 18:41:31
5	24ITR002	24ITT31	3	50	35	1	2026-10-26 18:41:31
6	24ITR002	24ITT32	3	60	43	1	2026-10-26 18:41:31
7	24ITR002	24ITP31	3	30	24	1	2026-10-26 18:41:31
8	24ITR002	24ITE31	3	60	44	1	2026-10-26 18:41:31
9	24ITR003	24ITT31	3	50	28	0	2026-10-26 18:41:31
10	24ITR003	24ITT32	3	60	34	0	2026-10-26 18:41:31
11	24ITR003	24ITP31	3	30	18	0	2026-10-26 18:41:31
12	24ITR003	24ITE31	3	60	32	0	2026-10-26 18:41:31
13	24CSR010	24CST31	3	60	55	0	2026-10-26 18:41:31
14	24ITR151	24ITT33	3	30	28	0	2026-10-26 18:56:26
15	24ITR153	24ITT96	3	37	30	0	2026-10-26 18:57:01
16	24ITR056	24ITT33	3	30	28	0	2026-10-26 18:59:41
17	24EER038	24ITT98	3	35	32	1	2026-10-26 19:02:07
18	24ECR029	24ITT97	3	37	36	0	2026-10-26 19:02:48
19	24CSR138	24ITT98	3	35	31	1	2026-10-26 19:05:04

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).

REFERENCES:

1. Oracle, Java SE Documentation – JDBC Overview and Architecture. https://docs.oracle.com/javase/8/docs/technotes/guides/jdbc/ .

2. Oracle, Java SE API – HttpServer. https://docs.oracle.com/en/java/javase/17/docs/api/jdk.httpserver/com/sun/net/httpserver/HttpServer.html .

3. Oracle, MySQL 8.0 Reference Manual. https://dev.mysql.com/doc/refman/8.0/en/ .

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5. University Grants Commission, Guidelines for Choice Based Credit System (CBCS). https://www.ugc.gov.in/pdfnews/8023719_Guidelines-for-CBCS.pdf .

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8. GeeksforGeeks, Establishing JDBC Connection in Java. https://www.geeksforgeeks.org/establishing-jdbc-connection-in-java/ .

9. YouTube, Java JDBC with MySQL Tutorial Videos. https://www.youtube.com/results?search_query=Java+JDBC+MySQL+tutorial .

10. Anthropic, Claude AI Assistant (used for project ideas and guidance). https://claude.ai/ .

GITHUB:

LINK:	**https://github.com/swathinagulsamy20-pixel/java1**